

Instruction

Homework 2 covers Chapter 3 to Chapter 4. Total 100 points.
Due by 8:00 AM May 22nd, 2026.
Submit your homework (hand-written or printed) in class, and please note that unclear formatting may negatively affect your score.
Language. We highly encourage you to write your answers in English.
Grading. Please briefly explain your answer. Answer without explanation will only receive half credit. **Plagiarism is strictly forbidden, and will receive zero credit for the entire homework (30% of this course) section.**
Should you have any question on this homework, you could i) ask in class, ii) post on canvas, or iii) post in class QQ group, and we will answer them as soon as possible.

Questions

1. Basic Functions and Interfaces (5 points)

Please describe the basic functions of the data link layer in detail, and explain its interface relationship with the physical layer and the network layer.

2. CRC Encoding Calculation (10 points)

Given the generator polynomial $G(x) = x^3 + x + 1$, perform CRC encoding on the data bit stream 1101011011. Calculate the checksum and provide the actual transmitted bit stream (with checksum).

3. Selective Repeat and GBN Protocols (10 points)

Provide simple schematic diagrams of the Selective Repeat (SR) and Go-Back-N (GBN) protocols, and briefly compare them in terms of their design goals and implementation details.

4. Stop-and-Wait Protocol Performance (5 points)

Given an ideal error-free point-to-point link with a bandwidth of 100 Mbps, a one-way propagation delay of 20 ms, and a frame length L of 1000 Bytes, calculate the channel utilization and maximum throughput of the Stop-and-Wait protocol.

计算机网络，2026年春季作业2

说明

作业2涵盖第3章至第4章，共100分。
截止时间为2026年5月22日上午8:00。
请在课堂上提交作业（手写或打印），并注意，格式不清晰可能会影响你的分数。

语言。我们强烈建议您用英语撰写答案。
评分。请简要解释您的答案。没有解释的答案将只能获得一半分数。抄袭是严格禁止的，并且将在整个作业（占本课程30%）部分获得零分。
整个家庭作业（占这门课程的30%）部分。
如果您对这次作业有任何问题，您可以 i) 在课堂上提问， ii) 在Canvas上发布，或 iii) 在课堂QQ群中发布，我们将尽快回答。

问题

1. 基本功能和接口（5分）

请详细描述数据链路层的基本功能，并解释其与物理层和网络层之间的接口关系。

2. CRC编码计算（10分）

给定生成多项式 $G(x) = x^3 + x + 1$ ，对数据比特流1101011011进行CRC编码。计算校验和，并提供实际传输的比特流（含校验和）。

3. 选择重发和GBN协议（10分）

提供选择重发（SR）和回退N（GBN）协议的简单示意图，并简要比较它们在设计目标和实现细节方面的异同。

4. 停等协议性能（5分）

给定一个理想的无误差点对点链路，带宽为100 Mbps，单向传播延迟为20ms，以及帧长度 L 为1000字节，计算停止等待协议的信道利用率和最大吞吐量。

5. CSMA/CD Working Principle (10 points)

Explain the working principle of CSMA/CD. Is it suitable for wireless networks? Give reasons for your answer.

6. Ethernet Collision Detection (10 points)

Consider an Ethernet using the CSMA/CD protocol. The bus segment length is 1 km, the signal propagation speed is 2×10^8 m/s, and the data rate is 1 Gbps.

1. Calculate the minimum frame length (in bits).
2. If stations A and B transmit data simultaneously and a collision occurs, what is the maximum time for A to detect the collision?
3. If the data frame transmitted by A is 1500 Bytes long, can it detect the collision before the transmission is completed? Please calculate and explain why or why not.

7. Switch Self-Learning and Forwarding (10 points)

A Fast Ethernet uses a switch to connect multiple hosts. The initial MAC address table of the switch is empty. Suppose Host A sends a frame to Host B, how does the switch forward it upon reception? Subsequently, Host B sends a frame to Host A, how does the switch forward it? After that, Host C sends a frame to Host A. Based on the above sequence, describe the self-learning process and forwarding decision of the switch.

8. IPv4 Datagram Fragmentation (10 points)

A host sends an IPv4 datagram with a total length of 4000 Bytes and a header length of 20 Bytes. It passes through a link with an MTU of 1500 Bytes. Please calculate:

1. How many fragments are needed? What is the total length of each fragment? What is the Fragment Offset of each fragment? And what is the MF bit of each fragment?
2. If the second fragment mentioned above is lost during transmission, can the receiver reassemble the datagram? What is the reason?

9. Shortest Path Calculation (10 points)

Consider a computer network topology as shown in the figure below (link weights represent network delays in milliseconds, ms):

5. CSMA/CD工作原理（10分）

解释CSMA/CD的工作原理。它适用于无线网络吗？请说明理由。

6. 以太网冲突检测（10分）

考虑一个使用CSMA/CD协议的以太网。总线段长度为1公里，信号传播速度为 2×10^8 米/秒，数据速率为1 Gbps。

1. 计算最小帧长（以比特为单位）。
2. 如果站点 A 和 B 同时传输数据且发生冲突，A 检测到冲突的最大时间是多少？
3. 如果 A 发送的数据帧长度为 1500 字节，它能在传输完成前检测到冲突吗？请计算并解释原因。

7. 交换机自学习和转发（10 分）

快速以太网使用交换机连接多个主机。交换机的初始 MAC 地址表为空。假设主机 A 向主机 B 发送帧，交换机在接收时如何转发？随后，主机 B 向主机 A 发送帧，交换机如何转发？之后，主机 C 向主机 A 发送帧。根据上述顺序，描述交换机的自学习过程和转发决策。

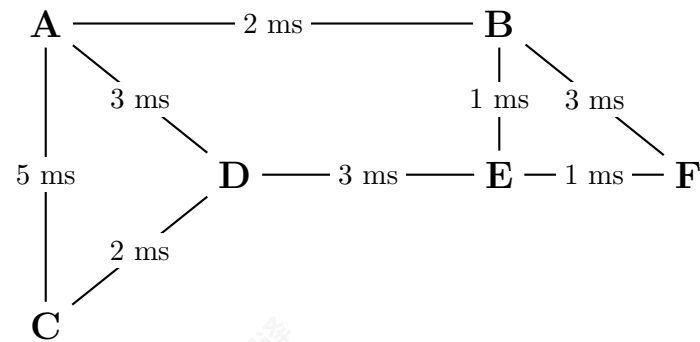
8. IPv4 数据报分片（10 分）

主机发送一个总长度为 4000 字节、头部长度为 20 字节的 IPv4 数据报。它通过一条 MTU 为 1500 字节的路由。请计算：

1. 需要多少个片段？每个片段的总长度是多少？
每个片段的 Fragment Offset 是什么？以及每个片段的 MF 位是什么？
2. 如果上述第二个片段在传输过程中丢失，接收器能否重新组装数据报？原因是什么？

9. 最短路径计算（10分）

考虑下图所示的计算机网络拓扑（链路权重代表网络延迟，单位为毫秒，ms）：



Please answer the following questions:

1. Using Dijkstra’s algorithm with node A as the source node, calculate the shortest paths from A to all other nodes (B, C, D, E, F) and their corresponding costs.
2. If the cost of link B-F is reduced from 3 ms to 1 ms, will the shortest path from node A to node F change? If it changes, what is the new shortest path and what is its cost?

10. IPv4 Subnetting (10 points)

Subnet the given network 192.168.10.0/24 into at least 6 subnets.
Provide: (1) The new prefix length; (2) The number of hosts in each subnet; (3) The detailed address range of each subnet and its hosts.

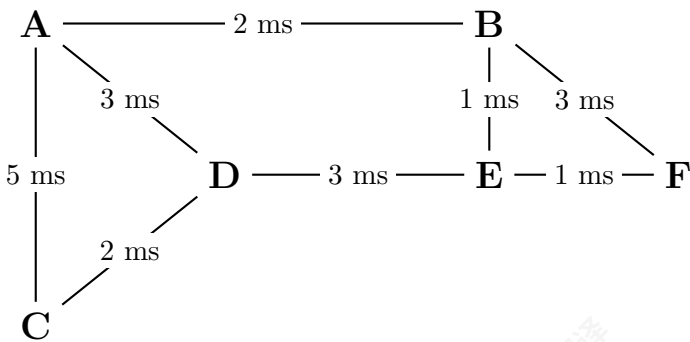
11. Router Forwarding Table (10 points)

A router has the following forwarding table:

Network Prefix	Next Hop
172.16.0.0/16	R1
172.16.8.0/21	R2
172.16.10.0/24	R3

Provide the forwarding results for the following addresses:

- 172.16.10.5
- 172.16.9.100
- 172.16.20.1



请回答以下问题：

1. 以节点A为源节点，使用迪杰斯特拉算法计算从A到所有其他节点（B、C、D、E、F）的最短路径及其对应成本。
2. 如果链路B-F的成本从3毫秒降低到1毫秒，从节点A到节点F的最短路径会改变吗？如果改变，新的最短路径是什么？成本是多少？

10. IPv4子网划分（10分）

将给定的网络192.168.10.0/24划分为至少6个子网。
提供：(1) 新的前缀长度；(2) 每个子网的主机数量；(3) 每个子网及其主机的详细地址范围。

11. 路由转发表（10分）

一个路由器具有以下转发表：

网络前缀	下一跳
172.16.0.0/16	R1
172.16.8.0/21	R2
172.16.10.0/24	R3

提供以下地址的转发结果：

- 172.16.10.5
- 172.16.9.100
- 172.16.20.1